```
>>>
    ========================================== RESTART: C:/Us
    Enter number of nodes: 6
    Enter number of edges: 7
    Enter source destination cost: 0 1 2
... 0 2 4
... 1 3 3
... 1 4 1
... 2 4 2
... 3 5 5
... 4 5 2
    Enter source destination cost: Enter source destination cost:
    urce destination cost: Enter start node: 0
    Enter goal node: 5
    Path: [0, 1, 4, 5]
    Cost: 5
>>>
```

1.py - C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py (3.14.1)

File Edit Format Run Options Window Help

```
n = int(input("Enter number of nodes: "))

graph = {}

for i in range(n):
    graph[i] = []

e = int(input("Enter number of edges: "))

for i in range(e):
    u, v, cost = map(int, input("Enter source destination cost: ").split())
    graph[u].append((v, cost))

start = int(input("Enter start node: "))
goal = int(input("Enter goal node: "))

pq = [(0, start, [start])]
visited = set()

while pq:
    cost, node, path = heapq.heappop(pq)

    if node == goal:
        print("Path:", path)
        print("Cost:", cost)
        break

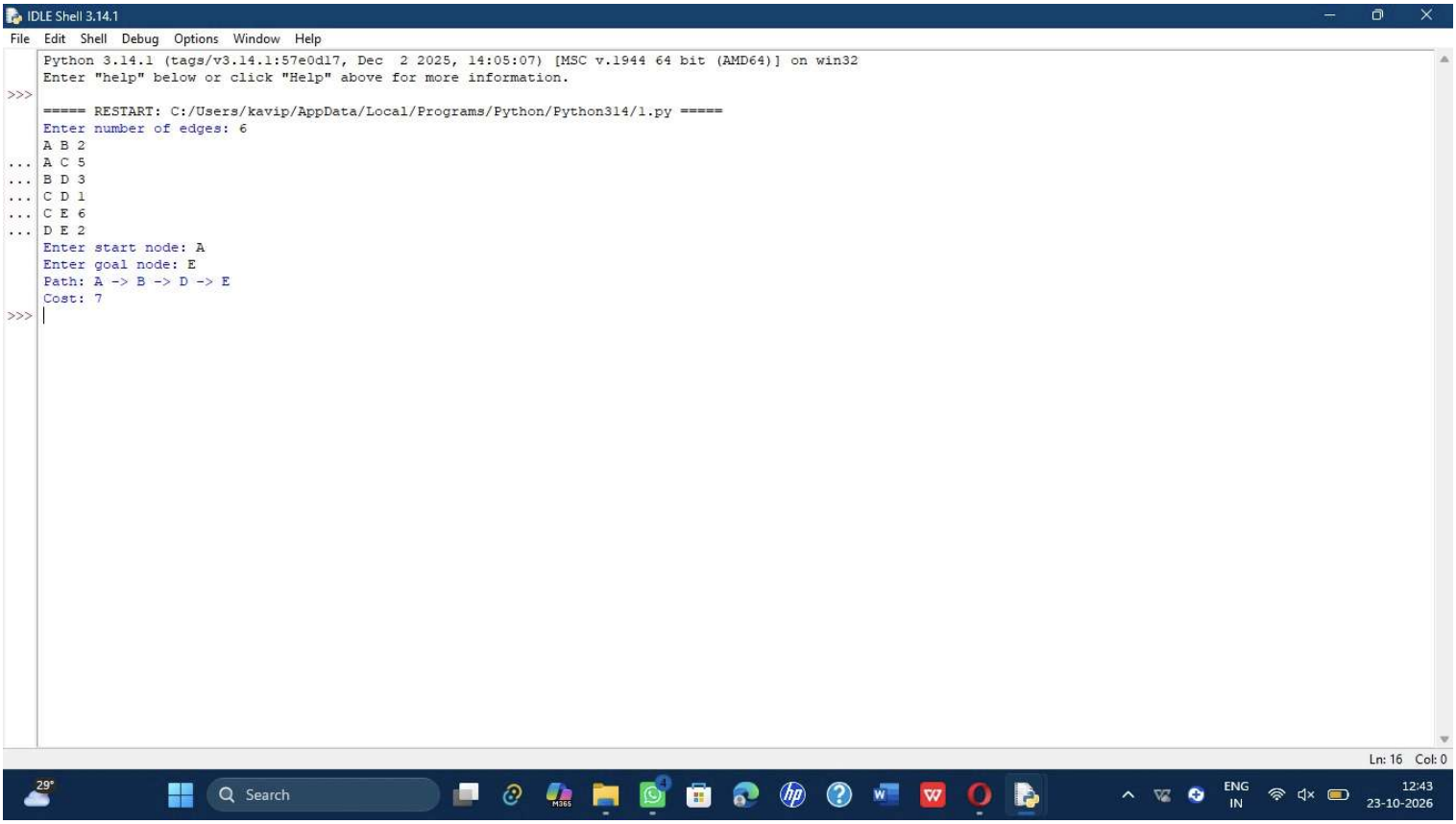
    if node in visited:
        continue

    visited.add(node)

    for neighbour, edge_cost in graph[node]:
        if neighbour not in visited:
            heapq.heappush(
                pq,
                (cost + edge_cost, neighbour, path + [neighbour])
            )
else:
    print("No path found")
```

Ln: 43 Col: 0

29° Search ENG IN 12:38 23-10-2026


IDLE Shell 3.14.1
File Edit Shell Debug Options Window Help
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec 2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Enter number of edges: 6
A B 2
... A C 5
... B D 3
... C D 1
... C E 6
... D E 2
Enter start node: A
Enter goal node: E
Path: A -> B -> D -> E
Cost: 7
>>>
Ln: 16 Col: 0
29°
Search
ENG
IN
12:43
23-10-2026